

Completeness and correctness of Čtrnák’s combinatorial enumeration of k -uniform tilings by regular polygons

Proof development (deliverable A; phase 2, round 1) — TilingAtlas working note

July 12, 2026

Status (updated after round 3: certificates + audit). The phase-1 skeleton was reviewed and approved on 2026-07-12; rounds 1–3 closed on the same date. **Every lemma of deliverable A is closed**, and round 3 discharged the machine and audit obligations: A3–A5 independently audited and re-run, A6 implemented and passing (44 letters pairwise non-isomorphic), P3 passing in both directions for $k \leq 6$, the deliverable-B implementation audit written (`audit-deliverable-B.md`: all eight hooks verified sound, seven fix-obligations recorded, none correctness-critical on valid input), and the historical $k = 8/9$ anomaly root-caused and closed by falsification (audit §4; hook 9 below). The classical inputs are down to two, both pinned: the Grünbaum–Shephard species table (Table 2.1.1, pp. 59–61, verified against the copy in `resources/papers`) and the Killing–Hopf classification of closed flat oriented surfaces. **0 lemmas are open.** Two structural changes came out of the closure work, both improvements: (1) round 1 corrected the flag dictionary ($\gamma = \sigma_2\sigma_0$, not σ_0 ; Remark 3.7); (2) round 2 replaced the planned orbifold-developability route through the C-block by the *direction bundle* (Remark 8.1), the translation-level cover that `eu_develop` literally iterates; all site symmetries unfold combinatorially, the developed object is a flat torus, and orbifold theory disappears from the proof entirely. Residual formalization debt is confined to three convention verifications flagged in Remark 8.12; each is finite bookkeeping, not a gap in the argument. Remaining risk now lives in deliverable B (does the C++ implement the abstract pipeline?) and in the unaudited machine certificates, exactly where a completeness claim about a program should keep its risk.

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1 Scope, conventions, and the theorem

1.1 Objects

Definition 1.1 (tiling, symmetry, uniformity). A *tiling* in this note is an edge-to-edge tiling T of the Euclidean plane $\mathbb{R}^2$ by regular polygons of edge length 1: a countable family of closed regular polygonal tiles with pairwise disjoint interiors whose union is $\mathbb{R}^2$, such that the intersection of any two tiles is empty, a common vertex, or a common full edge. $V(T)$, $E(T)$, $F(T)$ denote vertices, edges, faces. The *symmetry group* $G(T)$ is the group of all isometries of $\mathbb{R}^2$ (orientation-reversing ones included) mapping T to itself. T is *k-uniform* iff $G(T)$ has exactly k orbits on $V(T)$.

Convention 1.2 (chirality merge — definitional, not derived). Two tilings are *equivalent* iff some isometry of $\mathbb{R}^2$, orientation-reversing allowed, maps one to the other. Enumeration is up to this equivalence: a chiral tiling and its mirror image count once. This matches the convention of OEIS A068599 and the Galebach/Soto-Sánchez oracles, and is a settled repo decision. It is an axiom of the count, not a theorem.

Convention 1.3 (scope axioms). (i) Unit edge length is a normalization, not a restriction (a tiling by regular polygons of common edge length rescales). (ii) The polygon palette of the engine is $\mathcal{P} = \{3, 4, 6, 12\}$; the octagon is excluded from the palette and handled analytically by Lemmas O1–O4. (iii) $t1002$ denotes the truncated square tiling (4.8.8), the unique octagon-bearing tiling (Lemma O3); it is re-added to the $k = 1$ catalog by hand.

Definition 1.4 (the two tiling classes). For $k \geq 1$ let

$$\mathcal{T}_k = \{k\text{-uniform tilings containing no octagon}\}/\sim, \quad \mathcal{T}_k^{\text{all}} = \{\text{all } k\text{-uniform tilings}\}/\sim,$$

where $\sim$ is the equivalence of Convention 1.2. By Lemma O4, every member of $\mathcal{T}_k$ uses tiles from $\mathcal{P}$ only and has, after a rotation, all edge directions in the 30° grid; this is how the “12-direction” scope enters, as a consequence of octagon-freeness rather than as an extra hypothesis.

Remark 1.5 (statement hygiene, correcting the brief). The commissioning brief states the $k = 1$ case as $P_1 \cong \mathcal{T}_1 \setminus \{t1002\}$ with $\mathcal{T}_1$ already carrying the grid condition. That is redundant: with the grid (or octagon-free) condition in the definition, $t1002 \notin \mathcal{T}_1$ and the clean statement is $P_k \cong \mathcal{T}_k$ for *all* $k \geq 1$, with the octagon accounted for separately by Theorem 1.7. This note uses the clean split.

1.2 The abstract pipeline

The theorem is about the abstract algorithm specified in `reference/algorithm.txt`; deliverable B (the implementation-faithfulness audit) separately certifies that the C++ sources (`eu_solver`, `eu_pruner`, `eu_develop`) implement it. The abstract pipeline, over the alphabet $\mathcal{A}$ of Section 4, with target $k \leq \text{maxnum}$:

- Solve: the DFS of Section 6 over partial stub systems; emits every closed configuration that passes the rigidity filter (Section 7) with $1 \leq \# \text{vertices} \leq \text{maxnum}$.
- Prune: retains exactly one representative per isomorphism class of emitted configurations (Section 7).
- Develop: the geometric development map $M \mapsto \text{dev}(M)$ (Section 8).

Let P_k be the set of pruned outputs with exactly k vertices (k counting vertex types; for the regular palette every type is counting).

1.3 The theorem

Theorem 1.6 (main theorem: the engine). *Fix $k \geq 1$ and run the pipeline with $\text{maxnum} \geq k$. Then:*

(T) **Termination.** *The run halts.*

(S) **Soundness.** *For every $M \in P_k$, $\text{dev}(M)$ is defined and is a tiling whose class lies in $\mathcal{T}_k$; in particular its uniformity is exactly k .*

(C) **Completeness.** *For every class $[T] \in \mathcal{T}_k$ there is an $M \in P_k$ with $\text{dev}(M) \sim T$.*

(U) **Non-redundancy.** *The map $M \mapsto [\text{dev}(M)]$ is injective on P_k .*

Equivalently: dev induces a bijection $P_k \rightarrow \mathcal{T}_k$.

Theorem 1.7 (octagon scope). $\mathcal{T}_k^{\text{all}} = \mathcal{T}_k$ for every $k \geq 2$, and $\mathcal{T}_1^{\text{all}} = \mathcal{T}_1 \sqcup \{[t1002]\}$.

Corollary 1.8 (catalog completeness). *For $k \geq 2$ the developed catalog at k is in bijection with all k -uniform edge-to-edge tilings by regular polygons of common edge length, up to isometry with reflections; for $k = 1$ the catalog together with $t1002$ is.*

Remark 1.9 (evidence is not proof). The pipeline reproduces A068599 exactly for $k \leq 16$, with per-tiling agreement against the Soto-Sánchez data where available. Per the anti-fabrication discipline this is treated as evidence about the implementation and is used nowhere in the proof of Theorems 1.6 and 1.7. The one place the oracle legitimately enters is deliverable B, as a regression gate.

1.4 Obligation map

Obligation	Lemmas	Feeds
1 alphabet completeness	A1–A6	C (no missing letter), U (labels well defined)
2 local-rule soundness/necessity	L1, L2, S4	C (no wrongful prune), S (via C-block)
3 search exhaustiveness	T1, S1–S4	T, C
4 rigidity + dedup exactness	R1–R3, P1–P3	U, C, S
5 realizability	C1–C4	S
6 planar-quotient bridge	B0–B3	C, S
scope (octagon, grid)	O1–O4	Theorem 1.7
model well-definedness	D1a, D1b	everything

2 The octagon scope lemmas

Lemma 2.1 (O1: octagon-bearing species). [CLOSED] *In an edge-to-edge tiling by regular polygons, the multisets of interior angles at a vertex incident to an octagon, feasible by the angle equation alone, are exactly $\{4, 8, 8\}$ and $\{3, 8, 24\}$.*

Proof. Interior angles are $\theta_n = 180 - 360/n$ degrees, $\theta_n \geq 60$, and the angles at a vertex sum to 360. With $a \geq 1$ octagons ($\theta_8 = 135$): if $a = 2$ the remainder is 90, indivisible into two or more angles ≥ 60 , so it is a single square angle: $\{4, 8, 8\}$. If $a \geq 3$ the sum exceeds 360. If $a = 1$ the remainder is 225. As one angle: $225 > 180$, impossible. As two angles $\theta_m + \theta_n = 225$ with $\theta_m \leq \theta_n$ (so $\theta_m \leq 112.5$): the regular interior angles ≤ 112.5 are exactly $\theta_3 = 60$, $\theta_4 = 90$, $\theta_5 = 108$ (there is none strictly between 60 and 90, nor strictly between 90 and 108), and their complements to 225 are 165, 135, 117; of these only $165 = \theta_{24}$ is a regular angle ($135 = \theta_8$ re-enters the $a \geq 2$ case; $117 = 180 - 360/n$ has no integer solution), giving $\{3, 8, 24\}$. As three or more angles, each ≥ 60 : the sums attainable near 225 from $\{60, 90, 108, 120, \dots\}$ are $60+60+60 = 180$, $60+60+90 = 210$, $60+60+108 = 228$, $60+90+90 = 240$, and four angles give ≥ 240 , so 225 is never hit. Hence $\{3, 8, 24\}$ is the only $a = 1$ solution. $\square$

Lemma 2.2 (O2: 3.8.24 does not occur). [CLOSED] (*anchor*: [CLASSICAL: GRÜNBAUM–SHEPARD §2.1]) *No edge-to-edge tiling by regular polygons contains a vertex whose species is 3.8.24.*

Proof. Suppose vertex v carries tiles $\{T_3, T_8, T_{24}\}$. Consider the triangle T_3 , with vertices $v = v_0, v_1, v_2$. At v_0 , the two edges of T_3 separate it from T_8 and T_{24} in some order; say edge v_0v_1 borders the octagon. At v_1 the incident angles include T_3 's 60 and the octagon's 135; the remaining 165 must decompose into angles θ_n : as a single angle only $\theta_{24} = 165$ fits; as two or more, $165 - 60 = 105$ is not an angle and three angles exceed 165. So v_1 also has species 3.8.24, and the tile across the *other* edge of T_3 at v_1 , namely v_1v_2 , is a 24-gon. Repeating at v_2 : edge v_2v_0 borders an octagon. Now each vertex of T_3 must see one octagon and one 24-gon across the two incident triangle edges, i.e. the 3-cycle of T_3 's edges is properly 2-colored (octagon/24-gon) with colors alternating. A 3-cycle admits no such alternation, and specifically at v_0 both incident triangle edges now border octagons (v_0v_1 by choice, v_2v_0 by propagation), contradicting the species 3.8.24 at v_0 . (G&S state the exclusion of all six double-starred species, 3.8.24 included, in §2.1, p. 59 with Table 2.1.1; pinned against the copy in [resources/papers](#).) The proof above is self-contained regardless. $\square$

Lemma 2.3 (O3: an octagon forces $t1002$). [CLOSED] *Any edge-to-edge tiling by regular polygons containing at least one octagon is congruent to the truncated square tiling 4.8.8 ($t1002$). The*

truncated square tiling is 1-uniform; consequently no octagon occurs in any k -uniform tiling with $k \geq 2$.

Proof. Let T contain an octagon. By O1 and O2, every vertex of T incident to an octagon has species 4.8.8.

Step 1 (the vertex figure). At a vertex v of species 4.8.8 the cyclic tile order is (square, octagon, octagon). Hence: the two tiles adjacent to the square across its two edges at v are the two octagons; and each octagon has as cyclic neighbors at v the square on one side and the other octagon on the other.

Step 2 (squares adjacent to octagons). Let a square S share an edge e with an octagon. Both endpoints of e are octagon-incident, hence of species 4.8.8; by Step 1 the tiles across the two edges of S adjacent to e are octagons. Those octagons contain the endpoints of the edge e' of S opposite to e , so those endpoints are also of species 4.8.8, and Step 1 there makes the tile across e' an octagon too: all four neighbors of S are octagons, and all four vertices of S are of species 4.8.8. (In particular no two squares share an edge: a 4.8.8 vertex carries a single square.)

Step 3 (octagon coronas). Let O' be an octagon of T . Each of its vertices is 4.8.8, and by Step 1, at each vertex exactly one of the two edges of O' borders a square and the other an octagon. Consecutive edges of O' share a vertex, so the square/octagon assignment alternates around the eight edges; 8 is even, so the alternation is consistent, and the entire corona of O' is determined by the assignment at any single edge.

Step 4 (agreement induction). A copy of the truncated square tiling containing a given octagon is determined up to the corona parity of Step 3 (the point group of the model at an octagon center is D_4 , of index 2 in the D_8 of the octagon, and the two cosets realize the two parities); let M_0 be the copy containing O whose tile across one fixed edge e_0 of O matches T 's. Let A be the set of tiles t of T such that t is a tile of M_0 and every edge-neighbor of t in T equals the corresponding tile of M_0 . First, $O \in A$: its T -corona alternates (Step 3) with the same pinning as M_0 's (choice of M_0), and an octagon or square sharing a specified edge of O from the outside is determined as a set by its shape and that edge, so the corresponding tiles coincide. Now let $t \in A$ and let t' be the tile of T across an edge e of t ; then $t' \in M_0$ (membership of t in A), and we check $t' \in A$. If t' is an octagon, its T -corona alternates (Step 3) and is pinned by its neighbor t , which is also its M_0 -neighbor; determined-by-shape-and-edge again makes each of its eight neighbors equal to M_0 's. If t' is a square, then t is an octagon, by the structure of M_0 rather than Step 2: $t \in A$ makes t' the M_0 -neighbor of t across e , and in the truncated square tiling M_0 a square is edge-adjacent only to octagons, so a square neighbor t' forces t to be an octagon. Now t' is a square sharing edge e with the octagon t , so Step 2 applies to t' and makes all four T -neighbors of t' octagons; at each vertex v of the shared edge e , the third tile at v is the octagon completing the (4, 8, 8) figure, which shares a specified edge with t' , and the fourth neighbor shares the opposite edge; each is determined by shape and shared edge and so equals M_0 's tile. Hence A is closed under edge-adjacency; the tile-adjacency graph of a plane tiling is connected (join interior points of two tiles by a segment avoiding vertices), so every tile of T is a tile of M_0 , and a tiling contained tile-wise in another equals it: $T = M_0$.

1-uniformity. In the truncated square tiling the point group at an octagon center is D_4 and no vertex of the octagon lies on one of its mirrors (the mirrors pass through edge midpoints), so D_4 acts freely, hence transitively, on the octagon's 8 vertices; every vertex lies on an octagon, and all octagons are equivalent under translations. So the symmetry group is vertex-transitive: exactly one vertex orbit. $\square$

Lemma 2.4 (O4: octagon-free implies palette $\{3, 4, 6, 12\}$ and grid directions). [CLOSED] (*classical anchor: G&S Table 2.1.1, pp. 59–61*) *In an edge-to-edge tiling by regular polygons, only polygons with $n \in \{3, 4, 6, 8, 12\}$ occur. Hence an octagon-free tiling uses $\mathcal{P} = \{3, 4, 6, 12\}$ only; all its interior angles are multiples of 30° , and after a global rotation all edge directions lie in the 30° grid.*

Proof. The first sentence is classical, now pinned: G&S enumerate the 17 species (21 types counting cyclic order) of angle-feasible vertices in Table 2.1.1 and state on p. 59 that for the six species marked with two asterisks there, namely 3.7.42, 3.8.24, 3.9.18, 3.10.15, 4.5.20, 5.5.10, “there is no edge-to-edge tiling by regular polygons that includes even a single vertex of the species”; the surviving species use only $n \in \{3, 4, 6, 8, 12\}$. Each of the six exclusions is an odd-polygon adjacency argument of the O2 shape and can be reproved in an appendix if self-containedness is preferred. Glue: suppose the tiling is octagon-free, so all tiles lie in $\mathcal{P}$ and every interior angle is 60, 90, 120 or 150 degrees, a multiple of 30. Assign to each *edge-ray* (an edge directed away from one of its endpoints) its direction in $\mathbb{R}/360$. At a fixed vertex, consecutive rays in the rotational order differ by interior angles, so all rays at one vertex lie in a single coset of $30\mathbb{Z}$. The two rays of one edge differ by $180 = 6 \cdot 30$, so the coset is shared along every edge. The graph $(V(T), E(T))$ is connected (any two vertices are joined through the tile-adjacency chain along a segment between them, avoiding vertices by general position), so all edge-rays of T lie in one coset $\alpha + 30\mathbb{Z}$; rotating by $-\alpha$ places every direction on the grid. Finally, $t1002$ genuinely needs the 15° grid: its vertex figure mixes 90 and 135° degree angles, so its rays occupy a coset of $45\mathbb{Z}$ meeting no single coset of $30\mathbb{Z}$; this is why the truncated square tiling is outside the ζ_{12} model and outside $\mathcal{T}_1$. $\square$

Proof of Theorem 1.7 from O1–O4. A k -uniform tiling either contains an octagon, and is then $t1002$ with $k = 1$ (O3), or contains none and lies in $\mathcal{T}_k$ by definition. That $t1002$ is 1-uniform is part of the phase close of O3 (its group $p4m$ acts transitively on vertices). $\square$

3 The combinatorial model

This section carries the definitional groundwork (the risk-4 item) and is now closed: the stub-system axioms, the flag calculus of a tiling, the dictionary between the two, and the quotient construction, each with proof.

3.1 Stub systems

Definition 3.1 (stub system). A *stub system* is a tuple $(X, \rho, \mu, \gamma, c)$: a finite set X of *stubs*; a bijection $\rho : X \rightarrow X$ (**rneig**; $\lambda := \rho^{-1}$ is **lneig**); an involution $\mu : X \rightarrow X$ (**mirro**) with $\mu\rho\mu = \rho^{-1}$; a partial map $\gamma : X \rightharpoonup X$ (**glue**) with $\gamma^2 = \text{id}$ on its domain (fixed points allowed: self-gluing), domain μ -invariant and $\gamma\mu = \mu\gamma$ on the domain; and a coloring $c : X \rightarrow \mathcal{C}$ (**lvert**; for the regular palette $\mathcal{C} = \mathcal{P}$) satisfying $c \circ \mu = c \circ \rho$. A *vertex* of X is an orbit of $\langle \rho, \mu \rangle$. X is *closed* iff γ is total, *connected* iff $\langle \rho, \mu, \gamma \rangle$ acts transitively. Morphisms are maps commuting with ρ, μ, γ (where defined) and preserving c ; isomorphisms accordingly.

Remark 3.2 (parity is an axiom consequence). In any stub system, $\gamma(x)$ is μ -fixed iff x is: $\mu\gamma x = \gamma\mu x$, and γ is injective on its domain, so $\mu\gamma x = \gamma x \iff \mu x = x$. This is the abstract half of the mirror-parity rule (L2); no geometric input is needed.

Definition 3.3 (face walk, validity). The *face walk* is the partial map $\omega := \rho \circ \gamma$. A stub x with $n = c(x)$ satisfies: *condition 1* if c is constant along the ω -walk through x ; *condition 2a* if, when the walk through x is a cycle of minimal period ℓ , $\ell \mid n$; *condition 2b* if, when the walk is not a cycle, the maximal ω -chain through x has length $\leq n$ (length counts corners visited). A partial stub system is *valid* if all three hold at every stub where applicable.

Definition 3.4 (alphabet labeling). A stub system is *over the alphabet $\mathcal{A}$* (Section 4) if each of its vertices, as a substructure (orbit; ρ, μ, c) with γ forgotten, is isomorphic to an entry of $\mathcal{A}$. By Lemma A6 the entry is then unique, so vertices are canonically labeled.

Definition 3.5 (congruence, core, refinement). A *congruence* on a stub system is an equivalence relation θ on X , compatible with ρ, μ, γ ($x \theta y \Rightarrow f(x) \theta f(y)$, with γ -definedness constant on classes) and refining the fibers of c . The quotient X/θ is again a stub system. X is a *core* iff its only congruence is equality. The *refinement partition* $W(X)$ is the coarsest partition of X refining the c -fibers and stable under $\rho, \lambda, \mu, \gamma$; it is what `simplify` computes (Lemma R1).

3.2 Flags of a tiling and the dictionary

Definition 3.6 (flags). A *flag* of a tiling T is an incident triple $x = (v, e, f)$, vertex–edge–face. On $\text{Fl}(T)$ define the three involutions σ_0 (replace v by the other endpoint of e), σ_1 (replace e by the other edge of f at v), σ_2 (replace f by the other face at e). In an edge-to-edge tiling each is well defined and fixed-point free, and $\sigma_0\sigma_2 = \sigma_2\sigma_0$ (they alter disjoint components). The *dictionary* is

$$\rho := \sigma_2\sigma_1, \quad \mu := \sigma_2, \quad \gamma := \sigma_2\sigma_0, \quad c(x) := |\text{face of } \sigma_2 x|$$

(the color of a flag is the size of the *opposite* face across its edge). Every isometry in $G(T)$ commutes with $\sigma_0, \sigma_1, \sigma_2$ (they are incidence-defined) and preserves face sizes, hence commutes with the dictionary; orientation plays no role at the flag level, which is why flags rather than oriented darts are the right carrier and why Convention 1.2 is native to the model.

Remark 3.7 (correction to the phase-1 skeleton). Phase 1 declared $\gamma := \sigma_0$. That is wrong: with $\gamma = \sigma_0$ the face walk $\omega = \rho\gamma$ changes faces at every step and condition 1 would fail in $\text{Fl}(T)$ itself. The computation in D1a below forces $\gamma = \sigma_2\sigma_0$: when an edge is crossed, the local rotational sense reverses, and the extra σ_2 is that reversal. The error was caught by carrying out the walk computation rather than trusting the analogy with Delaney–Dress conventions; no lemma statement was affected.

Lemma 3.8 (D1a: flag calculus). [CLOSED] *Let T be a tiling. Then $(\text{Fl}(T); \rho, \mu, \gamma, c)$ satisfies the stub-system axioms with γ total, is connected, and is valid; moreover the ω -orbits are in bijection with the faces of T : the walk through a stub of color n is a cycle of length exactly n visiting the n corners of one face in rotational order.*

Proof. Axioms. $\mu^2 = \sigma_2^2 = \text{id}$. $\gamma^2 = \sigma_2\sigma_0\sigma_2\sigma_0 = \text{id}$ by commutation. $\gamma\mu = \sigma_2\sigma_0\sigma_2 = \sigma_2\sigma_2\sigma_0 \dots$ more directly $\gamma\mu = \sigma_2\sigma_0\sigma_2$ and $\mu\gamma = \sigma_2\sigma_2\sigma_0 = \sigma_0$; using commutation $\sigma_2\sigma_0\sigma_2 = \sigma_0\sigma_2\sigma_2 = \sigma_0$, so $\gamma\mu = \mu\gamma$. $\mu\rho\mu = \sigma_2(\sigma_2\sigma_1)\sigma_2 = \sigma_1\sigma_2 = (\sigma_2\sigma_1)^{-1} = \rho^{-1}$.

Local coordinates. Orient the plane once and for all. At a vertex v of degree d , list the incident edges $e_0, \dots, e_{d-1}$ in counterclockwise order (indices mod d) and let f_i be the face between e_i and e_{i+1} . The $2d$ flags at v are (v, e_i, f_i) and (v, e_i, f_{i-1}) ; write them $(i, 0)$ and $(i, 1)$. Direct

computation: $\sigma_1(v, e_i, f_i) = (v, e_{i+1}, f_i)$ and then σ_2 gives (v, e_{i+1}, f_{i+1}) , so $\rho(i, 0) = (i+1, 0)$; similarly $\rho(i, 1) = (i-1, 1)$; and $\mu(i, b) = (i, 1-b)$. Colors: $c(i, 0) = |f_{i-1}|$ and $c(i, 1) = |f_i|$, and one checks $c\mu = c\rho$ on both chirality types. (These formulas coincide with the generator's doubled dart structure; see A2.)

Crossing an edge. Let e carry index a at v and index b at the other endpoint w . Counterclockwise order at v appears clockwise from w , which gives the face matching $f_a^v = f_{b-1}^w$ and $f_{a-1}^v = f_b^w$. Hence $\sigma_0(v, e_a, f_a^v) = (w, e_a, f_{b-1}^w)$, a type-1 flag at w , and $\gamma(a, 0)_v = \sigma_2\sigma_0(v, e_a, f_a^v) = (w, e_a, f_b^w) = (b, 0)_w$: γ preserves chirality type relative to the global orientation.

Face walk. Take the corner $(\lambda x, x)$ with $x = (i+1, 0)$ at v ; its color is $c(x) = |f_i^v|$. Then $\gamma(x) = (j, 0)$ at w (the other end of e_{i+1} , with $f_i^v = f_j^w$ by the crossing rule), and $\omega(x) = \rho\gamma(x) = (j+1, 0)$ at w , whose color is $|f_j^w| = |f_i^v|$: the walk has advanced one corner around the same geometric face $f := f_i^v$ and kept its color. Iterating, the walk visits the corners of f in rotational order; since f is an embedded n -gon with n distinct corners, the walk first returns after exactly n steps. Conversely every corner of every face occurs in such a walk, giving the bijection. Conditions 1, 2a hold with $\ell = n$; there are no open chains (γ total). The same computation on type-1 flags gives the reverse walks; nothing changes.

Connectivity. Flags at one vertex are connected by ρ, μ . Any two vertices are joined by a path of edges (O4's connectivity argument), and γ crosses each edge. Hence $\langle \rho, \mu, \gamma \rangle$ is transitive. $\square$

Lemma 3.9 (B0: flag rigidity and discreteness). [CLOSED] *A symmetry of T fixing a flag is the identity; $G(T)$ acts freely on $\text{Fl}(T)$; the vertex set of T is uniformly discrete and $G(T)$ is a discrete group.*

Proof. Let g fix (v, e, f) : $g(v) = v$ and $g(e) = e$, so g fixes both endpoints of e , hence fixes the line of e pointwise; an isometry fixing a line pointwise is the identity or the reflection in that line, and the reflection swaps the two faces at e , contradicting $g(f) = f$. So $g = \text{id}$, and the action on flags is free. Uniform discreteness: two distinct vertices have distance ≥ 1 . Indeed if $0 < d(v, w) < 1$, then w lies within distance 1 of v ; every point at distance $< \frac{1}{2}$ from v lies in a tile incident to v (each unit-edge regular tile contains the disk sector of radius $\geq \sqrt{3}/2$ at each of its corners), and for $d(v, w) < 1$ one checks w lies on the boundary of a tile P incident to v ; by the edge-to-edge property a vertex of the tiling on ∂P is a corner of P , and distinct corners of a unit-edge regular polygon are at distance ≥ 1 from each other, a contradiction. Discreteness of $G(T)$: if $g_m \rightarrow \text{id}$, then for large m , g_m moves a fixed flag's vertex, edge midpoint and face barycenter by less than the minimal separation of those data, forcing g_m to fix that flag, so $g_m = \text{id}$. $\square$

Lemma 3.10 (D1b: quotients). [CLOSED] *Let $H \leq G(T)$ have finitely many orbits on $\text{Fl}(T)$. Then $T/H := (\text{Fl}(T)/H, \text{induced operations})$ is a finite closed connected stub system; its vertices correspond to H -orbits of $V(T)$. Moreover, for a stub $\bar{x} = Hx$ with $x = (v, e, f)$:*

- (i) $\mu\bar{x} = \bar{x}$ iff some $h \in H$ is the reflection in the line of e (a mirror along the edge, through both endpoints);
- (ii) $\gamma\bar{x} = \bar{x}$ iff some $h \in H$ is the rotation by π about the midpoint of e (the (a)-gluing);
- (iii) $\gamma\bar{x} = \mu\bar{x}$ iff some $h \in H$ is the reflection in the perpendicular bisector of e (the [a]-gluing).

Finally $T \mapsto T/H$ is functorial: an isometry g with $gHg^{-1} = H'$ induces an isomorphism $T/H \cong (gT)/H'$; in particular mirror-image tilings have isomorphic canonical quotients (chirality merge at the model level).

Proof. H commutes with the dictionary (Definition 3.6), so the operations descend; the axioms and closedness are inherited, connectivity passes to quotients, and finiteness is the orbit hypothesis. Vertices: $\langle \rho, \mu \rangle$ -orbits in the quotient are images of flag sets at single vertices, and two vertices' flag sets meet the same orbit iff the vertices are H -equivalent. For (i): $\mu\bar{x} = \bar{x}$ means $hx = \sigma_2x$ for some $h \in H$; such h fixes v and e , hence the line of e pointwise (as in B0), and is not the identity (the faces differ), so it is the reflection in that line; conversely that reflection realizes $hx = \sigma_2x$. For (ii): $hx = \sigma_2\sigma_0x$ maps (v, e, f) to (w, e, f') with $w \neq v$, $f' \neq f$: h preserves e , swaps its endpoints and swaps its sides, so h is the half-turn about the midpoint; conversely the half-turn does this. For (iii): $hx = \mu\gamma\bar{x}$ -wise, $hx = \sigma_0x$ maps (v, e, f) to (w, e, f) : h preserves e and f , swaps endpoints, so it fixes the midpoint and the perpendicular direction: the reflection in the perpendicular bisector. Functoriality: g maps flags of T to flags of gT commuting with everything, and descends to the orbit spaces. For the chirality statement take $H = G(T)$, g any reflection: $(gT)/G(gT) \cong T/G(T)$. $\square$

Remark 3.11 (presentation vs. structure: starred labels). The implementation presents each vertex orbit by choosing one “orientation sheet”: stubs of the chosen sheet are unstarred, their μ -images starred. The choice is arbitrary per orbit, and nothing in the structure depends on it; in particular γ may connect an unstarred stub to a starred one (the $[a\ b]$ Conway form) exactly when the two vertex orbits' chosen sheets disagree across the edge, since geometric γ preserves the *global* chirality type (D1a, crossing rule) while the sheets are local conventions. The four Conway bracket forms are therefore presentation data; the invariant content is D1b (i)–(iii). This is also why the pruner's isomorphism test, which never reads labels, is the right notion of sameness.

Lemma 3.12 (L1: necessity and heredity). [CLOSED] (i) *For any tiling T and $H \leq G(T)$ with finitely many flag orbits, T/H is valid: every ω -cycle has constant color n and length $\ell = n/r$ dividing n , where r is the order of the rotation part of the H -stabilizer of the corresponding face.* (ii) *Heredity: if M is closed and valid and P is a partial substructure of M (a subset of vertices carrying a subset of the gluings), then P is valid.*

Proof. (i) Fix a stub $\bar{x}$ of T/H and lift to $x \in \text{Fl}(T)$; by D1a the planar walk through x runs around a face f with $|f| = n = c(x)$, visiting flags $x, \omega x, \dots, \omega^{n-1}x$, all of color n ; colors descend, giving condition 1. The quotient walk closes at the minimal ℓ with $\omega^\ell x \in Hx$, say $hx = \omega^\ell x$ with $h \in H$; since H commutes with ω , h maps the walk orbit to itself and stabilizes f . *Claim: such h is orientation-preserving.* Define for a flag $y = (u, e, f)$ the local orientation $\text{or}(y) \in \{\pm 1\}$ as the orientation of the frame (direction of e away from u , direction from e into f). Each σ_i reverses or: the defining data of y and $\sigma_i y$ are exchanged by an explicit reflection (in the angle bisector of the corner for σ_1 , the perpendicular bisector of e for σ_0 , the line of e for σ_2), and or is covariant under isometries with the sign of their determinant. Hence $\omega = \sigma_2\sigma_1\sigma_2\sigma_0$ preserves or, so all flags in one ω -orbit share their value; from $\text{or}(hx) = \det(h)\text{or}(x)$ and hx in the orbit of x we get $\det(h) = +1$, proving the claim. So the return elements form the cyclic rotation group $C_r = \text{Stab}_H(f) \cap SO$, acting on the n corners of f freely with orbits of size r ; the minimal return is $\ell = n/r$, which divides n : condition 2a. Open chains do not occur (γ total). Reflections in $\text{Stab}_H(f)$, if any, do not shorten the cycle; they surface as coincidences between the cycle and its μ -image. (ii) Let $\iota : P \rightarrow M$ be the inclusion-style morphism (ρ, μ, c preserved, γ_P a restriction of γ_M). Walks in P are walks in M while defined and ι is injective, so: a P -cycle is an M -cycle with the same minimal period (deterministic walks return in P iff they return in M at the same step), giving 2a; a maximal open P -chain visits distinct stubs injectively inside an

M -cycle of length $\ell \leq n$, giving 2b; colors are inherited, giving 1. Monotonicity for the search: gluings are only ever added along a DFS branch, walks only extend, so a violation of 1/2a/2b in a configuration persists in all its descendants; equivalently no ancestor of a valid configuration violates the conditions. $\square$

Lemma 3.13 (L2: mirror parity). [CLOSED] *In any stub system, γ maps μ -fixed stubs to μ -fixed stubs (Remark 3.2); in T/H a stub is μ -fixed iff a mirror of H runs along its edge ($D1b(i)$), and the mirror line of an edge contains both endpoints, so the two end-stubs of such an edge are both μ -fixed. Consequently the solver's parity filter (glue self-mirrored only to self-mirrored) excludes no gluing that occurs in any T/H , and is forced for every closed valid system.*

Proof. All three assertions are contained in Remark 3.2 and $D1b(i)$; the last sentence restates them for the search. $\square$

Remark 3.14 (kinship with Delaney–Dress symbols). $Fl(T)/G(T)$ with $(\sigma_0, \sigma_1, \sigma_2)$ is the underlying set of the Delaney–Dress symbol of T ; the stub system repackages the operators as (ρ, μ, γ) and carries sizes as colors. The correspondence theorems (C, B) are vertex-centered analogues of the D-symbol realization theorems (Dress 1987, Dress–Huson 1987), cited as context and pattern only; the self-contained 2D proofs are shorter than a formal bridge between the conventions would be.

4 Obligation 1: the alphabet

The alphabet $\mathcal{A}$ (`mainlist`) is the finite set of vertex letters: pairs (configuration, site-symmetry variant), encoded as stub structures with the attachment data `ferkval/rebs`. The generator `alphabets/gen_alphabet.py` constructs $\mathcal{A}$ from first principles and gates it against the 44 legacy entries.

Lemma 4.1 (A1: configuration enumeration). [CLOSED] *The cyclic words over $\mathcal{P} = \{3, 4, 6, 12\}$ with interior angles summing to 360° , up to rotation and reflection, are exactly the following 14:*

$$\begin{array}{llll} (3, 12, 12), & (4, 6, 12), & (6, 6, 6), & (4, 4, 4, 4), \\ (3, 3, 6, 6), & (3, 6, 3, 6), & (3, 4, 4, 6), & (3, 4, 6, 4), \\ (3, 3, 4, 12), & (3, 4, 3, 12), & (3, 3, 3, 3, 6), & (3, 3, 3, 4, 4), \\ (3, 3, 4, 3, 4), & (3, 3, 3, 3, 3, 3), & & \end{array}$$

of which exactly $(4, 6, 12)$, $(3, 3, 4, 12)$ and $(3, 4, 4, 6)$ are chiral (not fixed by any reflection of the cyclic word).

Proof. With m polygons at a vertex, the angle equation reads $\sum_i (1 - 2/n_i) = 2$, i.e. $\sum_i 1/n_i = (m - 2)/2$ with $1/n_i \in \{1/3, 1/4, 1/6, 1/12\}$. Since each term is at most $1/3$, $m/3 \geq (m - 2)/2$ forces $m \leq 6$; since each term is positive and at most... for $m \leq 2$ no solution exists. Case $m = 3$ ($\sum = 1/2$): with a $1/3$: remaining $1/6$ from two terms: $1/12 + 1/12$ only, giving $\{3, 12, 12\}$; without: $1/4 + 1/6 + 1/12 = 1/2$ giving $\{4, 6, 12\}$, and $1/6 \cdot 3 = 1/2$ giving $\{6, 6, 6\}$; all other combinations miss. Case $m = 4$ ($\sum = 1$), by the number t of triangles: $t = 2$: remaining $1/3$ from two terms: $1/4 + 1/12$ ($\{3, 3, 4, 12\}$) or $1/6 + 1/6$ ($\{3, 3, 6, 6\}$); $t = 1$: remaining $2/3$ from three terms of $\{1/4, 1/6, 1/12\}$: only $1/4 + 1/4 + 1/6$ ($\{3, 4, 4, 6\}$); $t = 0$: only $1/4 \cdot 4$ ($\{4, 4, 4, 4\}$); $t \geq 3$ overshoots. Case $m = 5$ ($\sum = 3/2$): $t = 4$: fifth term $1/6$ ($\{3, 3, 3, 3, 6\}$); $t = 3$: two terms summing $1/2$: only $1/4 + 1/4$ ($\{3, 3, 3, 4, 4\}$); $t \leq 2$: maximum $2/3 + 3/4 < 3/2$.

Case $m = 6$: all six terms must be $1/3$ ($\{3^6\}$). Cyclic arrangements up to the dihedral action: 3-element multisets have one arrangement each; $\{3, 3, 6, 6\}$: equal pairs adjacent or alternating, 2; $\{3, 4, 4, 6\}$: the squares adjacent or separated, 2; $\{3, 3, 4, 12\}$: triangles adjacent or separated, 2 (adjacent: the two rotation classes $(3, 3, 4, 12)$ and $(3, 3, 12, 4)$ merge under reflection, leaving one chiral dihedral class); $\{4, 4, 4, 4\}$, $\{3, 3, 3, 3, 6\}$, $\{3^6\}$: one each; $\{3, 3, 3, 4, 4\}$: the lone triangle flanked by squares or not, i.e. $(3, 3, 3, 4, 4)$ and $(3, 3, 4, 3, 4)$, 2. Total $3 + 7 + 3 + 1 = 14$. Chirality: a cyclic word is achiral iff some rotation of its reversal equals it; checking each, exactly the three listed fail, e.g. the rotations of $(3, 3, 4, 12)$ are $(3, 3, 4, 12)$, $(3, 4, 12, 3)$, $(4, 12, 3, 3)$, $(12, 3, 3, 4)$ and its reversal is $(3, 3, 12, 4)$ up to rotation, not among them. (Consistency with the classical count: of G&S's 21 types over all polygon sizes, 15 occur in tilings; removing the type 4.8.8 leaves the 14 above.) $\square$

Lemma 4.2 (A2: variants are subgroup conjugacy classes). [CLOSED] *Fix a configuration w of length m and let $\text{Sym}(w) \leq D_m$ be the stabilizer of the cyclic word under the dihedral action on positions. For a tiling T and a vertex v with configuration w :*

- (i) *$\text{Stab}_{G(T)}(v)$ acts faithfully on the flags at v , and under the local coordinates of D1a this action is by word symmetries: rotations act as $(i, b) \mapsto (i + t, b)$ and reflections as $(i, b) \mapsto (a - i, 1 - b)$; the image is a subgroup $U \leq \text{Sym}(w)$.*
- (ii) *The vertex substructure of T/H at v (for any H , with $U = \text{image of } \text{Stab}_H(v)$) is the orbit space of the doubled dart structure of w under U , i.e. the generator's fold.*
- (iii) *Conjugate subgroups of $\text{Sym}(w)$ yield isomorphic folds; hence the map $\{\text{variants}\} \rightarrow \{\text{conjugacy classes of subgroups of } \text{Sym}(w)\}$ is a well-defined bijection onto, and no realizable vertex letter is missing from an alphabet that lists one fold per conjugacy class.*
- (iv) *(angle-weight clause, consumed by the C-block) With $a(n) := 6 - 12/n$, every ρ -cycle C of the fold of w by U has angle-weight sum $s(C) := \sum_{x \in C} a(c(x)) = 12/r$, where r is the order of the rotation subgroup of U ; in particular $s(C)$ divides 12.*

Proof. (i) An element h of the stabilizer fixes v , hence is a rotation about v or a reflection through v ; it permutes the incident edges preserving their counterclockwise cyclic order up to sense, and preserves the sizes of the faces between them, so its action on indices is an element of D_m stabilizing the word w ; the formulas for the two chirality types follow from D1a's coordinates (h orientation-preserving fixes each sheet, orientation-reversing swaps them). Faithfulness: h acting trivially on flags at v fixes a flag, hence $h = \text{id}$ by B0. (ii) The flags at v are the doubled dart set $\{(i, b)\}$ with the operations computed in D1a, which coincide with the generator's $\text{rneig}(i, 0) = (i+1, 0)$, $\text{rneig}(i, 1) = (i-1, 1)$, $\text{mirro}(i, b) = (i, 1-b)$, $\text{cls}(i, 0) = c[i-1]$, $\text{cls}(i, 1) = c[i]$; the quotient by $\text{Stab}_H(v)$ is by definition the fold by U . (iii) If $U' = sUs^{-1}$ with $s \in \text{Sym}(w)$, then s induces an isomorphism of folds $\text{darts}/U \rightarrow \text{darts}/U'$, $Ux \mapsto U'sx$, which respects the operations because s does. Surjectivity of the classification is (i): every stabilizer that occurs is a subgroup of $\text{Sym}(w)$, so its conjugacy class is listed. As a consistency check against the shipped table: $\text{Sym} = 1$ for the three chiral words (1 variant each), $\text{Sym} = \mathbb{Z}_2$ for the seven reflection-only words (2 variants), $\text{Sym}(3, 6, 3, 6) = D_2$ (5 subgroups, all normal: F, A1, A2, R, S), $\text{Sym}(6, 6, 6) = D_3$ (4 classes), $\text{Sym}(4, 4, 4, 4) = D_4$ (8 classes), $\text{Sym}(3^6) = D_6$ (10 classes; the two D_3 classes are the a/b pair, and the single D_2 class explains why $(3^6)S2$ has one edge-type axis and one triangle-type axis), for a total of $3 + 14 + 5 + 4 + 8 + 10 = 44$, the legacy entry count.

(iv) Lift a ρ -cycle of the fold to the doubled dart structure: ρ preserves the two sheets there, so the lifted walk runs within one sheet, stepping through the word positions in one rotational sense with corner weights $a(c_j)$, whose total over the whole word is 12 (the angle equation of A1, in 30° units). The folded cycle closes at the least t for which the lifted dart returns modulo U ; sheet-preserving elements of U are exactly its rotations, which form the cyclic group generated by the shift by m/r , so $t = m/r$. Rotations preserve the color word, so the color word is (m/r) -periodic and the partial weight sum over m/r consecutive corners is $12/r$. Sheet-1 cycles are the mirror images of sheet-0 cycles and carry the same weights. $\square$

Remark 4.3 (spurious variants are harmless). The brief asks that “none of the generated variants is spurious.” That direction is not load-bearing: a letter that never occurs in a valid closed core simply never appears in output, and soundness of what does appear is Obligation 5’s job, not the alphabet’s. Only missing letters (A2(iii)) and wrongly encoded letters (A3/A4) can break the theorem.

Lemma 4.4 (A3: table correctness). [MACHINE] (*implemented, unaudited*) *The generated (configuration $\times$ subgroup class) folds coincide, arrays and **ferkval** included, with the 44 legacy **mainlist** entries, trying every frame per fold; and the three consumed table copies (**solver_tables.inc**, **pruner_tables.inc**, the original hard-coded **mainlist**) agree.*

Lemma 4.5 (A4: structural identities). [MACHINE] (*implemented, unaudited*) *Every entry satisfies $\lambda\rho = \text{id}$, $\mu^2 = \text{id}$, $\mu\rho = \lambda\mu$, $c\mu = c\rho$.*

Lemma 4.6 (A5: the ferk transversal). [MACHINE] (*implemented, unaudited*) *For every entry D : $|\text{Aut}(D)| = \text{ferkval}(D)$, the action of $\text{Aut}(D)$ on stubs is free, and the generated **reps** list meets every $\text{Aut}(D)$ -orbit.*

Lemma 4.7 (A6: pairwise non-isomorphism — new certificate). [MACHINE] (**to be added**) *The 44 entries are pairwise non-isomorphic as colored stub structures (orbit; ρ, μ, c).*

Remark 4.8 (why A6 is needed, and that it was missing). The pruner buckets by the vertex-letter signature string and only compares within a bucket; the letter multiset must therefore be an isomorphism invariant of the underlying structure, which is exactly A6. Without it an over-count would be possible in principle: one tiling emitted under two letterings, landing in different buckets, kept twice. A6 is finite (44^2 colored-structure isomorphism tests) but is *not* among the generator’s current certificates; this note adds it as an obligation. Spot-check done by hand: the candidate collision (3, 6, 3, 6)A1 vs (3, 6, 3, 6)A2 is refuted by the corner colors at the μ -symmetric corners.

5 Obligation 2: local rules

L1 and L2 are stated and proved in Section 3 (Lemmas 3.12, 3.13), where they belong to the quotient analysis. What remains here is the scope remark:

Remark 5.1 (L3: sufficiency is the development theorem). The brief’s “conditions jointly sufficient for a closed configuration to be a valid combinatorial tiling” has no separate combinatorial content: the honest sufficiency statement is exactly the C-block (a closed valid rigid system develops to a genuine tiling), and it is proved there. Listing it here would double-count an obligation.

6 Obligation 3: search exhaustiveness

The abstract DFS. State: a connected partial stub system P over $\mathcal{A}$ with an ordered vertex list (letters indexed into $\mathcal{A}$). Root: a single fresh letter, no gluings. Step: pick a free stub s by any deterministic rule (the implementation uses `mincycle`, most-constrained-first, and normalizes s to the unstarred partner; the proofs below are indifferent to the rule); enumerate as children: (a) for every free stub t with the same μ -parity as s , the configuration $P + \{s \leftrightarrow t\}$, with the mirror pair $\mu s \leftrightarrow \mu t$ glued alongside when $s \neq \mu s$; the cases $t = s$ (self-gluing) and $t = \mu s$ are included; (b) if the vertex budget allows, for every letter $g \geq$ the root letter and every $r \in \text{reps}(g)$ of matching parity, the configuration $P + (\text{fresh vertex of letter } g) + \{s \leftrightarrow r\}$ (mirror pair alongside). Children failing conditions 1/2a/2b are discarded; closed children go to the rigidity filter; open children recurse.

Lemma 6.1 (T1: termination). [CLOSED] *The DFS tree is finite and the run halts.*

Proof. A configuration carries at most `maxnum` vertices of at most 12 stubs each (the largest letter, $(3^6)\text{F}$, has 12), so at most $12 \cdot \text{maxnum}$ stubs. Every child adds at least one gluing and none is ever removed along a branch, so branch length is at most the stub count. Branching at a node is bounded by the number of free stubs plus $|\mathcal{A}| \cdot \max_g |\text{reps}(g)| < \infty$. A finitely-branching tree with bounded branch length is finite; the root loop over the 44 letters is finite. $\square$

Lemma 6.2 (S1: guided descent — the no-drop core). [CLOSED] *Let M be a finite connected closed valid stub system over $\mathcal{A}$ with $m \leq \text{maxnum}$ vertices whose minimal letter is t^* . Then the DFS rooted at t^* visits at least one closed configuration isomorphic to M .*

Proof. Say a configuration P follows M via ψ if $\psi : X_P \rightarrow X_M$ is injective, preserves ρ, μ, c , and carries each gluing of P to a gluing of M (so $\gamma_M \psi = \psi \gamma_P$ on $\text{dom } \gamma_P$). Since ψ preserves ρ and μ and letters are single $\langle \rho, \mu \rangle$ -orbits, ψ maps each vertex of P bijectively onto a vertex of M of the same letter, and distinct vertices to distinct vertices.

Base. M has a vertex of letter t^* ; the root configuration (one fresh t^* , no gluings) follows M via any letter isomorphism onto it, which exists by Definition 3.4.

Inductive step. Suppose the DFS visits a non-closed P that follows M via ψ ; let s be the free stub the rule picks. M is closed, so $y := \gamma_M(\psi s)$ exists. By Remark 3.2 applied in M , y is μ -fixed iff ψs is, iff s is (ψ preserves μ): parities match.

Case (a): $y = \psi(t)$ for some $t \in X_P$. Then t is free in P : otherwise $\psi \gamma_P(t) = \gamma_M(y) = \psi s$ with ψ injective would give $\gamma_P(t) = s$, contradicting s free. The child $P' := P + \{s \leftrightarrow t\}$ (mirror pair alongside if $s \neq \mu s$) is among the enumerated children, and ψ still follows: the new gluings map to $\{\psi s \leftrightarrow y\}$ and, for the mirror pair, to $\{\mu \psi s \leftrightarrow \mu y\}$, which is a gluing of M because γ_M commutes with μ .

Case (b): y 's vertex is not in the image of ψ . Its letter g satisfies $g \geq t^*$ because t^* is minimal among M 's letters, and P has fewer than $m \leq \text{maxnum}$ vertices (its vertices inject into M 's and miss y 's), so the fresh-vertex branch is enabled and tries letter g . The set of letter isomorphisms $D_g \rightarrow (y\text{'s vertex in } M)$ is a nonempty torsor under $\text{Aut}(D_g)$, so the stubs r with some isomorphism sending $r \mapsto y$ form exactly one $\text{Aut}(D_g)$ -orbit; by A5 the `reps` list meets it, say at r_0 with isomorphism ψ_0 , $\psi_0(r_0) = y$. Parity of r_0 matches (ψ_0 preserves μ -fixedness and y 's parity matches s 's). The child $P' := P + (\text{fresh } g) + \{s \leftrightarrow r_0\}$ is enumerated, and $\psi' := \psi \cup \psi_0$ follows M (ψ_0 's image vertex is new, so injectivity holds; the gluing maps correctly as in case (a)).

No prune. A configuration following M is isomorphic (via ψ , onto its image) to a partial substructure of M , hence valid by L1(ii); so **checkpart** never discards a child produced above.

Closure. Each step adds a gluing of M 's finite gluing set, so after finitely many steps the guided branch reaches a closed P^* following M via some ψ^* . Its image is closed under ρ, μ (unions of vertices) and under γ_M (P^* closed and $\gamma_M \psi^* = \psi^* \gamma_{P^*}$), hence equals X_M by connectedness of M ; an injective surjective morphism is an isomorphism. Since the DFS explores every child of every visited node and the root is visited, every node of the guided branch is visited, in particular P^* . $\square$

Lemma 6.3 (S2: minimal-letter rooting; sharding). [CLOSED] *The run roots one DFS at every letter $i \in \mathcal{A}$. The search rooted at i visits exactly (up to isomorphism) the closed valid systems over $\mathcal{A}$ whose minimal letter is i ; in particular every M as in S1 is reached from its minimal letter, and the root sets of distinct roots are disjoint. Sharding the root loop (`EU_SHARD_N/W`) partitions the roots; the union of the shards' outputs equals the sequential run's output.*

Proof. Containment one way: the fresh-vertex branch only adds letters $g \geq \text{vertype}[0] = i$ and the root letter i is never removed, so every configuration in the tree rooted at i has letter multiset with minimum exactly i . Containment the other way is S1 applied with root $t^*(M) = i$. Disjointness: the minimal letter is an isomorphism invariant. Sharding assigns each root to exactly one worker and workers share no state, so the multiset union of emissions over workers equals the sequential emission multiset; the pruner's output is invariant under this repartition because its dedup is by isomorphism class within k (P1/P2), which is insensitive to emission order across families. (The last step's implementation correctness, byte-identity of the merged catalog, is an audit item; the mathematical statement here is set equality of the pruned classes.) $\square$

Remark 6.4 (correcting the brief's description). The brief describes a “nondecreasing-vertex-type tie-break.” The code constrains new letters only by $g \geq \text{vertype}[0]$ (min-root invariant, `eu_solver.cpp:680`); letters may otherwise arrive in any order. This is weaker than nondecreasing order and makes S1/S2 easier, since the descent never needs to reorder M 's vertices.

Lemma 6.5 (S3: emission gates). [CLOSED] (*abstract*) + *audit-facing* *In the abstract pipeline, every closed configuration visited with $1 \leq \#vertices \leq \text{maxnum}$ that passes the rigidity filter is emitted. On the regular palette no budget, cap, or filter other than conditions 1/2a/2b and the rigidity filter gates emission.*

Proof. Abstractly this is the definition of Solve (Section 1); there is nothing to prove. The content is entirely implementation-side and is carried by audit hooks 4 and 7: `seen = 0` emits every $k \in [1, \text{maxnum}]$; the noncounting budget is inert on the regular palette (`has_noncounting` false); `EU_STREAM` changes the sink only; `EU_KONLY` is a user-selected per- k restriction applied before dedup and does not affect the selected k . $\square$

Lemma 6.6 (S4: the picked stub exists; the local tests are the conditions). [CLOSED] (*abstract*) + *audit-facing* *In a non-closed configuration every free stub lies on an open ω -walk, so a most-constrained free stub is well defined; and the abstract DFS prunes exactly on conditions 1/2a/2b. The choice rule affects traversal order only: S1's proof is indifferent to which free stub is picked.*

Proof. A free stub x has $\gamma(x)$ undefined, so the maximal ω -chain ending at ρ -predecessors of x is open, and x bounds it; hence open walks exist iff free stubs exist. The second and third assertions restate the abstract pipeline definition and S1. That **checkpart** decides exactly

1/2a/2b (walking each cycle from every starting stub, so every maximal open chain is checked at full length from its earliest stub) and that `mincycle` returns a free stub are implementation facts: audit hook 5. $\square$

7 Obligation 4: rigidity filter and dedup

Lemma 7.1 (R1: refinement computes the coarsest congruence). [CLOSED] *Let M be a finite closed stub system. Then: (i) M has a coarsest congruence $\theta_{\max}$, and it equals the stable refinement partition $W(M)$ of Definition 3.5; (ii) M is a core iff $W(M)$ is discrete; (iii) both implementations of `simplify` (the solver’s iterated sort-refinement and the pruner’s pair-elimination) compute $W(M)$, so `simplify` accepts M iff M is a core.*

Proof. (i) A partition is *stable* iff whenever x, y share a class, $c(x) = c(y)$ and each of $\rho, \lambda, \mu, \gamma$ maps their classes into a common class. For partitions into equivalence relations this stability is verbatim the compatibility defining a congruence, because all four operations are (total, on a closed system) functions: every congruence is a stable partition refining the c -fibers, and every stable partition refining the c -fibers is a congruence. The coarsest stable partition refining a given initial partition exists and is computed by iterated refinement (standard partition-refinement theory; Cardon–Crochemore, Paige–Tarjan), which is $W(M)$; hence $W(M) = \theta_{\max}$. (ii) is immediate from (i). (iii) The solver’s `simplify` is the textbook iterated refinement on the 6-tuple $(c, \text{class}, \text{class} \circ \mu, \text{class} \circ \gamma, \text{class} \circ \lambda, \text{class} \circ \rho)$ and converges to the coarsest stable partition. The pruner’s computes the greatest symmetric relation R below the complete relation that is compatible with colors and the four operations, by deleting violating pairs to a fixpoint. Diagonal pairs never violate (their tests compare a pair to itself), so $\Delta \subseteq R$; compatible symmetric relations containing Δ are closed under composition of relations, and the transitive closure of R is again compatible and below the complete relation, so by maximality R is transitive, hence an equivalence, hence (by (i)) a congruence, so $R \subseteq \theta_{\max}$; conversely $\theta_{\max}$ is a compatible symmetric relation below the complete one, so $\theta_{\max} \subseteq R$ by maximality. Thus $R = W(M)$. (That the two code paths implement these two computations faithfully is audit hook 6.) $\square$

Remark 7.2 (why Weisfeiler–Leman anxiety does not apply). Color refinement is famously incomplete for isomorphism and automorphism detection on general graphs. Stub systems are not general graphs: their relations are *functions* (deterministic transition structures), and on those, refinement computes the coarsest congruence exactly; this is Myhill–Nerode/DFA-minimization logic, as `algorithm.txt` itself hints by calling the check “DFA minimization.” Both directions of the brief’s obligation-4 worry (a rigid map wrongly rejected; two distinct maps wrongly merged) are settled by R1 and P1 respectively.

Corollary 7.3 (R2: rejection drops nothing). [CLOSED] *If a closed valid M fails the filter (M not a core), the tiling $\text{dev}(M)$ is still enumerated: its canonical quotient $T/G(T)$, $T := \text{dev}(M)$, is a core (B2b), is reached independently (S1–S2), passes the filter (R1), and develops back to T (B3(a)). Hence discarding non-cores never removes a tiling from the catalog.*

Corollary 7.4 (R3: accepted solutions have exact uniformity). [CLOSED] *If M is a core with m vertices, then $G(\text{dev}(M)) = H(M)$ and $\text{Fl}(\text{dev}(M))/G(\text{dev}(M)) \cong M$; in particular $\text{dev}(M)$ is exactly m -uniform. (Proof inside B3(c): a strictly larger symmetry group would induce a proper congruence on M .)*

Lemma 7.5 (P1: pruner exactness on cores). [CLOSED] *Let M, M' be finite connected closed stub systems that are cores. Then `comparesolutions`(M, M') returns true iff $M \cong M'$.*

Proof. The procedure computes, by deleting violating pairs to a fixpoint, the greatest symmetric relation R on $X \sqcup X'$ that is contained in the initial relation $\Delta \cup (X \times X') \cup (X' \times X)$ and compatible with colors and the four operations. As in R1, Δ survives, and the procedure returns true iff some class of R is not a singleton, i.e. iff R contains a cross pair.

($\Leftarrow$) If $\varphi : M \rightarrow M'$ is an isomorphism, the relation $\Delta \cup \text{graph}(\varphi) \cup \text{graph}(\varphi^{-1})$ is symmetric, compatible, and inside the initial relation; an easy induction shows the deletion process never removes a pair of a compatible relation all of whose operation-image pairs it also contains, so R retains the cross pairs of φ and the procedure returns true.

($\Rightarrow$) Suppose R contains a cross pair and set $R' := R \cap (X \times X')$, a nonempty relation compatible componentwise with colors and operations (the fixpoint conditions transfer each cross pair's operation-images into R'). The composite $S := R'^{\top} \circ R' \subseteq X' \times X'$ is symmetric, compatible, and contains the diagonal of the range of R' ; the equivalence it generates together with $\Delta_{X'}$ is compatible (compatibility survives symmetric closure, union with Δ , and transitive closure, because the operations are functions), hence is a congruence on M' ; M' a core forces it to be equality, so R' is single-valued. Symmetrically, M a core forces R' injective. The domain of R' is closed under $\rho, \lambda, \mu, \gamma$ (images of pairs are pairs), hence is all of X by connectedness; symmetrically R' is surjective. A total, injective, surjective, color- and operation-preserving relation that is single-valued is an isomorphism.

Both core hypotheses are supplied by the pipeline ordering invariant: `simplify` runs before `comparesolutions` on every block, in file mode and in `EU_STREAM` mode alike. This ordering is a correctness precondition, not an optimization; audit hook 6 pins it. $\square$

Lemma 7.6 (P2: bucketing never separates duplicates). [CLOSED] *The bucket key (letter-signature string, refinement fingerprint) is an isomorphism invariant, so isomorphic solutions land in the same bucket and are compared; within a bucket every new solution is compared against all kept ones. Hence, granting P1, exactly one representative per isomorphism class survives.*

Proof. The signature is the multiset of vertex letters; an isomorphism maps vertices to vertices preserving (ρ, μ, c) -structure, hence (by A6) preserves letters, hence the multiset and its canonical string. The fingerprint is a hash of the sorted multiset of per-stub colors after three rounds of refinement seeded by c ; each round's recoloring is defined from isomorphism-invariant data only, so isomorphic systems produce equal multisets and equal hashes. (Hash collisions between non-isomorphic systems merely place them in one bucket, where P1 separates them; nothing is lost.) The bucket therefore contains every previously kept isomorph, and `compareToSeen` tests against all of them. $\square$

Lemma 7.7 (P3: geometric cross-check via canonical form N). [MACHINE] *(optional, cheap) For pruned $M \neq M'$ at the same k : $N(\text{symbol}(\text{dev}(M))) \neq N(\text{symbol}(\text{dev}(M')))$, and for every solver-emitted duplicate pair merged by the pruner the N -keys agree; N is the proven canonical form of `docs/canonical-form/`.*

Remark 7.8 (relation to canonical form N; answering the brief directly). The pipeline computes no canonical form and does not need one: the pruner decides isomorphism pairwise, and P1 proves that decision exact on cores. Canonical form N lives at the geometric level and is related by the chain: combinatorial isomorphism of cores $\iff$ congruence of developed tilings (B3) $\iff$ equal N -keys (N 's Corollary 5.5). This makes N an independent, already-proven arbiter

that can machine-audit every merge and non-merge on real runs (P3). If P1 had resisted closure, replacing the pruner verdict by N -key equality at develop time was the designed escape hatch; with P1 closed, P3 is belt and braces.

8 Obligation 5: realizability (development)

Remark 8.1 (architecture: the direction bundle replaces orbifolds). Phase 1 planned this block through orbifold theory: realize M as a compact flat 2-orbifold, quote developability (Thurston’s orbifold chapter), pull back. That plan is sound but imports machinery (orbifold gluings with mirror boundary, orbifold coverings) whose fine print a referee must re-verify. The closed proofs below use a lighter route, and it is the route `eu_develop` literally computes: the *direction bundle* $B(M)$, the cover of M that unfolds every site symmetry by remembering an edge direction in $\mathbb{Z}/12$. Cone points and mirrors disappear upstairs: $B(M)$ assembles a closed flat *oriented* surface, necessarily a torus, and the only classical input left is the classification of such surfaces. Orbifolds remain the right mental picture of the quotient; they no longer occur in any proof.

Definition 8.2 (angle weights; property $(\angle)$). For $n \in \mathcal{P}$ let $a(n) := 6 - 12/n \in \{2, 3, 4, 5\}$, the interior angle of the regular unit n -gon in units of 30° . A closed valid stub system has property $(\angle)$ if every ρ -cycle C satisfies $s(C) := \sum_{x \in C} a(c(x)) \mid 12$. By A2(iv), every stub system over $\mathcal{A}$ has $(\angle)$.

Lemma 8.3 (C0: congruence quotients preserve the hypotheses). [CLOSED] *If M is finite, connected, closed, valid with $(\angle)$ and θ is a congruence on M , then M/θ is finite, connected, closed, valid with $(\angle)$.*

Proof. All structure descends (Definition 3.5); finiteness, connectedness and closedness are immediate. Validity: the quotient map q intertwines ω , so the walk through $q(x)$ is the q -image of the walk through x ; its minimal period ℓ' divides the original ℓ , colors are constant as images of constants, and $\ell' \mid \ell \mid n$ gives condition 2a; open chains do not occur. $(\angle)$: a ρ -cycle C' of M/θ is the image of a ρ -cycle C with $|C'|$ dividing $|C|$; the color word of C is $|C'|$ -periodic, so $s(C') = s(C) |C'|/|C|$, which divides $s(C)$, hence 12. $\square$

Definition 8.4 (the direction bundle). Let M be finite, connected, closed, valid with $(\angle)$. Since μ normalizes $\langle \rho, \gamma \rangle$, the $\langle \rho, \gamma \rangle$ -orbits on X are either one ($X^+ := X$) or two swapped by μ ($X = X^+ \sqcup \mu X^+$; M is then called *ultrachiral*, the solver’s own term). Fix X^+ and a base stub $x_0 \in X^+$. The *direction bundle* is $B(M) := X^+ \times \mathbb{Z}/12$ with two families of steps:

$$\text{corner steps } (x, d) \rightarrow (\rho x, d + a(c(\rho x))), \quad \text{edge steps } (x, d) \rightarrow (\gamma x, d + 6).$$

B_0 denotes the connected component of $(x_0, 0)$. Elements (x, d) are *darts*; *bundle edges* are the pairs $\{(x, d), (\gamma x, d+6)\}$; the *face walk* is $\hat{\omega} := (\text{edge step, then corner step})$. The *corner* between a dart and its corner-step image is assigned the color $c(\rho x)$ and the angle $a(c(\rho x))$, matching the corner convention of D1a.

Lemma 8.5 (C1: the bundle surface). [CLOSED] (i) *The corner-step cycle through (x, d) projects onto the ρ -cycle $C \ni x$, traverses it $12/s(C)$ times, and carries total angle weight exactly 12.* (ii) *Every face walk is a cycle; the walk through a dart of color n has length exactly n and constant color n .* (iii) *Every dart lies on exactly one face walk, and each bundle edge is traversed once by*

each of the face walks through its two darts, in opposite directions. (iv) Gluing one closed unit-edge regular n -gon per face walk of B_0 along the bundle edges (isometrically on edges, reversing the boundary walk directions) yields a compact, connected, flat, oriented surface $S(B_0)$.

Proof. (i) After $|C|$ corner steps the direction has advanced by $s(C)$; the cycle closes at the least t with $ts(C) \equiv 0 \pmod{12}$, namely $t = 12/s(C)$, an integer by $(\angle)$; the total weight is $ts(C) = 12$. (ii) One $\hat{\omega}$ -step from a dart of color n advances the direction by $6 + a(n) \equiv -12/n \pmod{12}$ and projects to one ω -step of M . The underlying M -walk is a cycle of some length $\ell \mid n$ with constant color n (validity), so the bundle walk can close only at multiples $t\ell$; the direction returns first at $t\ell \cdot 12/n \equiv 0 \pmod{12}$, i.e. $t = n/\ell$: the bundle walk is a cycle of length exactly n . (iii) $\hat{\omega}$ is a bijection on darts, so its orbits partition them. The walk through (x, d) traverses the bundle edge $\{(x, d), (\gamma x, d+6)\}$ exactly when it leaves (x, d) ; the walk through the partner leaves $(\gamma x, d+6)$, crossing the same edge with direction labels differing by 6, i.e. oppositely. (iv) The identification space of finitely many closed polygons under a complete isometric pairing of boundary edges is a compact surface; its points are face-interior points, edge-interior points (two flat half-disks), and vertex points. At a vertex point, the corners glued around it are enumerated by alternately crossing a glued edge and moving to the adjacent corner of the next polygon; under the conventions of Definition 8.4 this alternation is the corner step (the corner between the edges of (x, d) and of its corner-step image is a corner of the polygon of color $c(\rho x)$ glued along both), so by (i) the link is a single cycle of total angle 12 units $= 2\pi$: a flat disk. Hence $S(B_0)$ is flat everywhere. Orienting every polygon positively along its face walk, (iii) says each gluing reverses boundary orientations, so the face orientations assemble to a global orientation. Compactness is finiteness of B_0 ; connectedness is connectedness of B_0 . $\square$

Lemma 8.6 (C2: classification). [CLASSICAL: KILLING–HOPF; E.G. WOLF, SPACES OF CONSTANT CURVATURE] *A closed flat oriented surface is isometric to $\mathbb{C}/\Lambda$ for a lattice $\Lambda = \mathbb{Z}t_1 + \mathbb{Z}t_2 \subset \mathbb{C}$ of rank 2, acting by translations.*

Proof note. Classical. The elementary route: a compact flat surface is complete, so its universal cover is the Euclidean plane (Killing–Hopf); deck transformations are fixed-point-free isometries, orientation-preserving by orientability of the quotient; a fixed-point-free orientation-preserving planar isometry is a translation; compactness forces rank 2. $\square$

Definition 8.7 (development; the marked model). Fix an isometry $S(B_0) \cong \mathbb{C}/\Lambda$ and let $p : \mathbb{C} \rightarrow S(B_0)$ be the covering map. $\text{dev}(M) := \hat{T}$ is the tiling of $\mathbb{C}$ whose tiles are the closures of the connected lifts of the open faces of $S(B_0)$. Normalize the identification (rotating, and conjugating if needed) so that the base dart $(x_0, 0)$ develops with edge direction 0 and corner steps rotate positively. A *walk flag* of $\hat{T}$ is a flag reading off a lifted face-walk traversal; $F^\circ(\hat{T})$ denotes the walk flags (one of the two orientation classes of $\text{Fl}(\hat{T})$). The *marked model* is the bijection

$$F^\circ(\hat{T}) \cong \{(b, z) : b \in B_0, z \in \text{pos}(b) + \Lambda\},$$

where pos is one fixed development of positions, under which corner steps fix z and the edge step at a dart with direction label d adds ζ^d , with $\zeta := e^{i\pi/6}$. $\Phi : F^\circ \rightarrow X^+$ is the stub projection $(b, z) \mapsto (\text{stub of } b)$.

Lemma 8.8 (C3a: the developed tiling). [CLOSED] *$\hat{T}$ is an edge-to-edge, Λ -periodic tiling of $\mathbb{C}$ by unit-edge regular polygons with colors in $\mathcal{P}$ and all edge directions on the 30° grid: the edge of every dart (x, d) develops in direction $d \cdot 30^\circ$. The marked model is well defined, and it is rigid:*

a position assignment $B_0 \rightarrow \mathbb{C}$ compatible with the step rules is determined by its value at one dart. Φ commutes with ρ and γ and preserves colors. Tiles do not overlap and leave no gaps by construction (the plane is the cover of an intrinsic gluing, not the target of an immersion), which answers the brief's overlap worry and is why this block will transfer to non-convex tiles unchanged. The exact area identity $\sum \text{face areas} = |\det(t_1, t_2)|$ remains a runtime evidence gate, never a proof step.

Proof. Faces of $S(B_0)$ have simply connected interiors, so each lifts isometrically to $\mathbb{C}$; the closures are unit-edge regular n -gons; they have disjoint interiors and cover $\mathbb{C}$ because the faces decompose $S(B_0)$ and p is a covering; edge-to-edge because the identifications pair full unit edges; Λ -periodic because Λ is the deck group. Directions: the flat metric on $\mathbb{C}/\Lambda$ has trivial holonomy, so every edge of $S(B_0)$ has a well-defined direction in $\mathbb{C}$; assigning each dart its developed direction, a corner step rotates by the interior angle $a \cdot 30^\circ$ positively (normalization) and an edge step reverses (+6 units), matching the bundle labels; by induction from the base dart along the connected B_0 , the label d is the direction in units. The same induction gives rigidity of the marked model: positions propagate deterministically along corner steps (fixed) and edge steps (add ζ^d), and B_0 is connected. Φ 's properties are read off the construction. $\square$

Lemma 8.9 (C3b: affine realization of bundle maps). [CLOSED] *Let M_1, M_2 be as in Definition 8.4, with developments $\hat{T}_1, \hat{T}_2$ and marked models. Let $f : X_1^+ \rightarrow X_2^+$ be a surjection and $\varepsilon \in \{+1, -1\}$, $h \in \mathbb{Z}/12$, such that $\beta(x, d) := (f(x), \varepsilon d + h)$ maps steps of $B(M_1)$ to steps of $B(M_2)$ (for $\varepsilon = -1$: to reversed steps); explicitly, for $\varepsilon = +1$ this holds iff f commutes with ρ, γ and preserves colors, and for $\varepsilon = -1$ iff $f\rho = \rho^{-1}f$, $f\gamma = \gamma f$ and $c \circ f = c \circ \rho$. Assume $\beta(B_0^{(1)}) \subseteq B_0^{(2)}$ (arrange by composing with a direction shift). Then for every choice of a base pair there is a unique isometry*

$$g(z) = \zeta^h z + w \quad (\varepsilon = +1), \quad g(z) = \zeta^h \bar{z} + w \quad (\varepsilon = -1)$$

such that $(b, z) \mapsto (\beta b, g z)$ maps walk flags of $\hat{T}_1$ to walk flags of $\hat{T}_2$; consequently $g(\hat{T}_1) = \hat{T}_2$.

Proof. Step compatibility of (β, g) : corner steps fix z on both sides; the edge step at (x, d) adds ζ^d , its β -image is the edge step at $(fx, \varepsilon d + h)$ adding $\zeta^{\varepsilon d + h}$, and indeed $g(z + \zeta^d) = g(z) + \zeta^h \zeta^d = g(z) + \zeta^{d+h}$ for $\varepsilon = 1$, respectively $g(z + \zeta^d) = g(z) + \zeta^h \bar{\zeta}^d = g(z) + \zeta^{h-d}$ for $\varepsilon = -1$. (The hypothesis on colors is what makes β map corner steps correctly: for $\varepsilon = -1$, the reversed corner step out of $f\rho x = \rho^{-1}fx$ adds $a(c(\rho\rho^{-1}fx)) = a(c(fx)) = a(c(\rho x))$, using $c \circ f = c \circ \rho$; this is exactly the axiom $c\mu = c\rho$ when $f = \mu$.) Pin w by one base pair; by rigidity (C3a) the position assignment $b \mapsto g(\text{pos}_1(\beta^{-1}\text{-compatible base}))$ agrees with pos_2 up to the model's Λ_2 -freedom, so (β, g) maps every walk flag of $\hat{T}_1$ to a walk flag of $\hat{T}_2$. Faces: by C1(ii) every face walk in either bundle has length exactly its color, so β restricted to a face walk is a bijection onto a face walk of the same color, and g maps the corresponding developed tile isometrically onto a developed tile. Since f is onto, every tile of $\hat{T}_2$ arises, so $g(\hat{T}_1) \subseteq \hat{T}_2$ tile-wise between two tilings, forcing equality. Uniqueness: an isometry is determined by the image of one flag. $\square$

Lemma 8.10 (C3c: M is the quotient of its development). [CLOSED] *Define $\hat{\Phi} : \text{Fl}(\hat{T}) \rightarrow X$ by $\hat{\Phi} := \Phi$ on walk flags and $\hat{\Phi} := \mu \circ \Phi \circ \sigma_2$ on the other orientation class, and set*

$$H(M) := \{ g \in \text{Isom}(\mathbb{C}) : g(\hat{T}) = \hat{T} \text{ and } \hat{\Phi} \circ g = \hat{\Phi} \}.$$

Then $\hat{\Phi}$ is a surjective morphism of stub systems, $H(M) \leq G(\hat{T})$ acts freely on flags with finitely many orbits, its orbits are exactly the $\hat{\Phi}$ -fibers, and $\hat{\Phi}$ descends to an isomorphism

$\text{Fl}(\hat{T})/H(M) \cong M$. Moreover $H(M)$ contains orientation-reversing elements iff M is not ultrachiral.

Proof. $\hat{\Phi}$ is a morphism. On walk flags this is C3a. On the other class, with $\tilde{y}$ a walk flag: colors, $c(\sigma_2\tilde{y}) = |\text{face}(\tilde{y})| = |\text{face}(\sigma_1\tilde{y})| = c_{\hat{T}}(\rho\tilde{y}) = c(\Phi\rho\tilde{y}) = c(\rho\Phi\tilde{y}) = c(\mu\Phi\tilde{y}) = c(\hat{\Phi}\sigma_2\tilde{y})$; ρ : $\rho\sigma_2 = \sigma_2\rho^{-1}$, so $\hat{\Phi}(\rho\sigma_2\tilde{y}) = \mu\Phi\rho^{-1}\tilde{y} = \rho\mu\Phi\tilde{y} = \rho\hat{\Phi}(\sigma_2\tilde{y})$; γ commutes with σ_2 ; and $\hat{\Phi}\circ\sigma_2 = \mu\circ\hat{\Phi}$ on both classes, so $\hat{\Phi}$ intertwines the mirrors. Surjectivity: onto X^+ by C3a; onto μX^+ (which is X^- or X^+) via the second clause.

Freeness. An isometry fixing a flag is the identity (B0 applies to $\hat{T}$).

Orbits = fibers. $H(M)$ preserves fibers by definition. Transitivity, walk-flag part: two walk flags over the same stub x are $(b_1, z_1), (b_2, z_2)$ with $b_i = (x, d_i)$; the direction shift $\delta(y, e) := (y, e + d_2 - d_1)$ is a β -map of C3b with $\varepsilon = 1, f = \text{id}$, preserving B_0 (it maps the component of b_1 to that of b_2 , both B_0); realize it by $g(z) = \zeta^{d_2-d_1}z + w$ pinned by $(b_1, z_1) \mapsto (b_2, z_2)$: then $g \in H(M)$ and g connects the two flags. Cross part: if M is not ultrachiral, μ maps X^+ to itself and satisfies the $\varepsilon = -1$ hypotheses of C3b (the stub-system axioms verbatim), so after a direction-shift adjustment it is realized by some orientation-reversing g_- with $\hat{\Phi} \circ g_- = \hat{\Phi}$ (checked on walk flags: $\hat{\Phi}(g_-\tilde{y}) = \mu\Phi(\sigma_2g_-\tilde{y}) = \mu\mu\Phi\tilde{y} = \Phi\tilde{y}$, since σ_2g_- is the flag realization of μ); g_- maps the walk-flag part of a fiber to the other part, and $H(M) = H^+ \sqcup H^+g_-$ acts transitively on each fiber. If M is ultrachiral, fibers over X^+ contain walk flags only (the second clause of $\hat{\Phi}$ lands in $X^- = \mu X^+$), transitivity is the walk-flag part, and no orientation-reversing g can preserve $\hat{\Phi}$ (it would force Φ -values in X^-).

Quotient. A surjective morphism whose fibers are the orbits of a freely acting group descends to an isomorphism from the orbit space; finiteness of the orbit set is $|X| < \infty$. $\square$

Lemma 8.11 (C3d: round trip and functoriality). [CLOSED] (i) If T is a tiling and $H \leq G(T)$ has finitely many flag orbits, then $\text{dev}(T/H)$ is congruent to T ; normalizing $\text{dev}(T/H) = T$, one has $H(T/H) = H$. (ii) If θ is a congruence on M and $q : M \rightarrow M/\theta$ the quotient, the developments can be normalized so that $\text{dev}(M) = \text{dev}(M/\theta)$ as tilings of $\mathbb{C}$, $\hat{\Phi}_{M/\theta} = q \circ \hat{\Phi}_M$, and $H(M) \leq H(M/\theta)$. (iii) If $M_1 \cong M_2$ then $\text{dev}(M_1) \sim \text{dev}(M_2)$.

Proof. (i) T 's own flags supply a step-compatible position assignment for $B_0(T/H)$: send a walk flag of T to (its image stub and direction, its vertex position); the step equations hold in T by the geometry of its tiles. Rotate T so the base flag has direction 0; by rigidity (C3a) this assignment coincides with the marked model, so the developed flag set of $\text{dev}(T/H)$ equals the flag set of T : the tilings are equal. $H(T/H) \supseteq H$ is immediate (H preserves T and the quotient labeling). For the converse let $g \in H(T/H)$: g preserves every fiber of $\text{Fl}(T) \rightarrow \text{Fl}(T)/H$, i.e. every H -orbit; picking a flag $\tilde{y}$, there is $h \in H$ with $g\tilde{y} = h\tilde{y}$, and $h^{-1}g$ fixes a flag, so $g = h \in H$ by freeness (B0). (ii) $\beta_q(x, d) := (qx, d)$ maps steps to steps (colors descend; C0 gives the hypotheses on M/θ), and bases can be chosen compatibly; C3b with $\varepsilon = 1, h = 0$ realizes it by $g = \text{id}$ once the two developments are normalized with matching base positions: $\text{dev}(M) = \text{dev}(M/\theta)$ literally, and $\hat{\Phi}_{M/\theta} = q \circ \hat{\Phi}_M$ by construction, whence every isometry preserving $\hat{\Phi}_M$ -fibers preserves the coarser $\hat{\Phi}_{M/\theta}$ -fibers. (iii) An isomorphism satisfies the $\varepsilon = 1$ hypotheses of C3b; the realized g carries $\hat{T}_1$ to $\hat{T}_2$. $\square$

Remark 8.12 (the referee surface). Three places above compress convention verifications into a sentence; each is finite bookkeeping, flagged here per the anti-fabrication discipline. (1) C1(iv): the identification of the vertex-link alternation with the corner step (the corner between the edges of a dart and of its corner-step image belongs to the polygon glued along both). (2)

C3a: the positivity convention for corner steps under the chosen orientation (fixed by allowing a conjugation in the normalization). (3) C3c: the color computation on the non-walk orientation class, written out in the proof but relying on the D1a convention $c(y) = |\text{face}(\sigma_2 y)|$. All three are checkable mechanically from Definitions 3.6 and 8.4.

Lemma 8.13 (C4: `eu_develop` computes the marked development). [CLOSED] *(abstract) + audit-facing* On input a finite connected closed valid M with $(\angle)$, the BFS of `eu_develop` computes a position assignment for B_0 compatible with the step rules (its *star* walk enumerates corner-step cycles; its glue step is the edge step); it terminates within the guards; the recorded position discrepancies generate exactly Λ ; the integer elimination plus Lagrange–Gauss reduction output a basis of Λ ; and the seed reduction outputs exactly one representative position per vertex of $\hat{T}$ modulo Λ . The developer therefore succeeds on every valid input; in the pipeline it only ever runs on cores, for which Λ is the full translation lattice of $G(\text{dev}(M))$ (R3), so the emitted cell is the true period cell.

Proof. The algorithm’s keys are pairs (stub, direction) and its two moves are precisely the corner and edge steps, so the explored key set is B_0 and the computed positions are a step-compatible assignment; by rigidity (C3a) it is *the* development up to the base normalization. Termination and guards: a corner-step cycle has length equal to the degree of the developed vertex, at most 6, within the 60-step guard; the number of keys is at most $12|X| \leq 144\text{maxnum}$, within the 200k pop guard for every reachable target (audit hook 8 converts both guards into assertions with these bounds). Discrepancies: the BFS implicitly chooses a spanning tree of the step graph of B_0 (the tree of first placements); positions are propagated along tree steps, and every non-tree step compares a propagated position against an existing one, recording the difference, which lies in Λ because both are positions of the same B_0 -point in the marked model. These differences generate Λ : the step graph of B_0 is the vertex-edge graph of the torus tiling of $S(B_0)$ enriched with direction data, its fundamental cycles surject onto π_1 of the surface (attaching the face disks only adds relations), and the discrepancy of a cycle is exactly its Λ -holonomy. The integer elimination (a Hermite-style column reduction) and Lagrange–Gauss steps compute a basis of the generated lattice by standard integer linear algebra, exactly (float enters only the reduction coefficient by rounding, never position identity). The seed pass reduces the placed vertex positions modulo the basis and dedupes to one representative per vertex class: with *exact* coset reduction (the canonical representative of Def. 2.2 of the N -note, two integer floor divisions) this holds unconditionally. The shipped code instead reduces via float barycentric coordinates, which agrees with the exact reduction whenever every developed coordinate is bounded by $12|X| \leq 144\text{maxnum}$ (each edge step adds a unit ζ^d and there are $\leq 12|X|$ steps): the reachable range keeps every coordinate small (the committed golden $k \leq 6$ catalog has all reduced seed and lattice-basis coordinates ≤ 14 ; instrumenting the pre-reduction developing map gives ≤ 8), against a float break-point near 10^7 where a translated congruent copy first mis-reduces — five to six orders of margin — and *fix-obligation FB-1* of the audit replaces the float reduction by the exact one to remove the caveat entirely. $\square$

9 Obligation 6: the planar–quotient bridge

B_0 is proved in Section 3 (Lemma 3.9).

Lemma 9.1 (B1: k -uniform implies cocompact wallpaper, hence periodic). [CLOSED] *modulo citation pin* ([CLASSICAL: BIEBERBACH; G&S §1.4/3.5 SECONDARY]) *If T is k -uniform then*

$G(T)$ has at most $12k$ orbits on flags and acts cocompactly on $\mathbb{R}^2$; being discrete (B0) and cocompact, it is one of the 17 wallpaper groups and contains a rank-2 lattice of translations. Corollary: k -uniform implies doubly periodic.

Proof. Flags at a vertex number $2d \leq 12$ ($d \leq 6$ since all angles are $\geq 60^\circ$), and flags over one vertex orbit form at most 12 flag orbits, so at most $12k$ in all. Cocompactness: let $v_1, \dots, v_k$ represent the vertex orbits and let R be the circumradius of the dodecagon; every point of $\mathbb{R}^2$ lies in some tile, hence within R of some vertex, hence within R of gv_i for some $g \in G(T)$, $i \leq k$; so the compact set $\bigcup_i \bar{B}(v_i, R)$ meets every orbit of points, i.e. the action is cocompact. The classification of discrete cocompact planar isometry groups and the existence of the translation lattice is Bieberbach's theorem in dimension 2 (citation to pin; any standard crystallographic reference suffices). $\square$

Lemma 9.2 (B2a: the canonical quotient is a valid stub system over $\mathcal{A}$). [CLOSED] For $[T] \in \mathcal{T}_k$, $M := T/G(T)$ is a finite connected closed valid stub system over $\mathcal{A}$ with exactly k vertices.

Proof. Finiteness, closedness, connectedness and the vertex correspondence are D1b with $H = G(T)$ and B1 (finitely many flag orbits); the vertex count is k by definition of k -uniformity. Validity is L1(i). Over the alphabet: each vertex substructure is a fold of its configuration word by a subgroup of the word's symmetry group (A2(i)–(ii)); the configuration is one of A1's fourteen (O4 restricts sizes to $\mathcal{P}$, and the angle equation holds at a tiling vertex); the alphabet lists one fold per conjugacy class (A2(iii), certified against the table by A3), and conjugate folds are isomorphic, so the substructure is isomorphic to an entry. $\square$

Lemma 9.3 (B2b: the canonical quotient is a core). [CLOSED] For a tiling T with finitely many flag orbits, $T/G(T)$ is a core.

Proof. Let θ be a congruence on $M := T/G(T)$. By C3d(i) normalize $\text{dev}(M) = T$ with $H(M) = G(T)$. By C3d(ii), $H' := H(M/\theta)$ satisfies $G(T) = H(M) \leq H'$, and H' preserves $\text{dev}(M/\theta) = \text{dev}(M) = T$, so $H' \leq G(T)$, forcing $H' = G(T)$. By C3c applied to M/θ , the fibers of $\hat{\Phi}_{M/\theta} = q \circ \hat{\Phi}_M$ are the H' -orbits = the $G(T)$ -orbits = the fibers of $\hat{\Phi}_M$; hence q is injective and θ is equality. $\square$

Lemma 9.4 (B3: the Galois correspondence and the bijection). [CLOSED] Let T be a tiling and $H \leq G(T)$ a subgroup with finitely many flag orbits; normalize $\text{dev}(T/H) = T$ (C3d(i)).

- (a) (Round trip.) $\text{Fl}(\text{dev}(M))/H(M) \cong M$ for every finite connected closed valid M with $(\angle)$ (C3c), and $\text{dev}(T/H) \sim T$ (C3d(i)).
- (b) (Correspondence.) The assignments $H' \mapsto \theta_{H'} := (\text{the partition of } \text{Fl}(T)/H \text{ induced by } H' \text{-orbits})$ and $\theta \mapsto H((T/H)/\theta)$ are mutually inverse, inclusion-preserving bijections between the intermediate groups $H \leq H' \leq G(T)$ and the congruences on T/H .
- (c) (Bijection.) $M \mapsto [\text{dev}(M)]$ and $[T'] \mapsto T'/G(T')$ are mutually inverse bijections between isomorphism classes of cores over $\mathcal{A}$ with k vertices and $\mathcal{T}_k$. In particular every tiling has exactly one core, namely $T'/G(T')$.

Proof. (a) is C3c and C3d(i).

(b) Forward map well defined: H' normalizes nothing special but commutes with the flag operations and preserves colors, so its orbit partition on $\text{Fl}(T)$ descends to a congruence on

$\text{Fl}(T)/H$ (classes are H' -orbits, unions of H -orbits since $H \leq H'$). *Injective*: suppose $\theta_{H'} = \theta_{H''}$, i.e. H' and H'' have the same flag orbits. For $h' \in H'$ and a flag $\tilde{y}$ there is $h'' \in H''$ with $h'\tilde{y} = h''\tilde{y}$; then $h''^{-1}h'$ fixes a flag, so $h' = h'' \in H''$ (B0), giving $H' \subseteq H''$ and symmetrically equality. *Backward map lands in the interval*: for a congruence θ on $M := T/H$, C3d(ii) gives $H \leq H(M/\theta)$ and $H(M/\theta)$ preserves $\text{dev}(M/\theta) = \text{dev}(M) = T$, so $H(M/\theta) \leq G(T)$. *Backward then forward is the identity*: the fibers of $\hat{\Phi}_{M/\theta} = q \circ \hat{\Phi}_M$ (C3d(ii)) are, by C3c applied to M/θ , the $H(M/\theta)$ -orbits; their induced partition on $M = \text{Fl}(T)/H$ is the fiber partition of q , which is θ . *Forward then backward is the identity*: for $H \leq H' \leq G(T)$, the quotient $M/\theta_{H'}$ is canonically $\text{Fl}(T)/H'$, and $H(\text{Fl}(T)/H') = H'$ by the argument of C3d(i) (an isometry preserving every H' -orbit lies in H' , by freeness). *Inclusion-preserving*: clear in both directions.

(c) Well defined: if M is a core with k vertices, then applying (b) to $T' := \text{dev}(M)$, $H := H(M)$ (using $M \cong T'/H(M)$ from (a)): the congruence $\theta_{G(T')}$ is proper iff $G(T') \supsetneq H(M)$; a core admits no proper congruence, so $G(T') = H(M)$, the uniformity of T' is the number of $H(M)$ -vertex-orbits = the number of vertices of M , namely k ; T' is octagon-free because its tiles carry colors in $\mathcal{P}$; so $[T'] \in \mathcal{T}_k$. Isomorphic cores develop to congruent tilings (C3d(iii)), and congruent tilings have isomorphic canonical quotients (D1b functoriality), so both maps are well defined on classes. Round trips: $\text{dev}(T'/G(T')) \sim T'$ is C3d(i); $T'/G(T') \cong M$ for $T' = \text{dev}(M)$, M a core, is (a) together with $G(T') = H(M)$ just shown. B2a/B2b put $T'/G(T')$ in the right codomain. Hence mutually inverse bijections, and the core of a tiling is unique. $\square$

9.1 Verdict on the brief's obligation-6 sharpening

Adopted, with corrections. The brief's analysis is right on the two points that matter, and both were verified rather than assumed:

1. *No period bound is needed anywhere.* The enumerated object is the quotient itself: a solution with k vertex letters has at most $12k$ stubs, so finiteness of the search (T1) is bare combinatorics, and no bound on the translation lattice of the developed tiling enters T, S, C, or U. A period bound is what a *geometric* enumerator needs; demanding one here would import the mixed-type finiteness problem that stalled the lattice method, for no benefit. Datta–Maity (arXiv:1705.05236) and Kundu–Maity (arXiv:2111.15484) are demoted to context citations; independently, their bounds are per-Archimedean-type and would not cover the mixed-type case anyway.
2. *The split into 6a/6b is correct.* 6a (=B1) is classical: discreteness (B0) plus cocompactness plus Bieberbach; “ k -uniform $\Rightarrow$ periodic” falls out. 6b is the real content and is exactly B2 + B3 here.

Corrections and refinements to the brief's formulation: (i) discreteness of $G(T)$ must be established before Bieberbach applies (B0). (ii) The quotient is in general a flat *orbifold* with possible mirror boundary, not a torus; but the closed proof does not run in the orbifold category at all: the direction bundle (Remark 8.1) passes to the translation-level cover, where the object *is* a torus, and the site symmetries return as the explicit affine maps of C3b/C3c. The brief's instinct (“torus”) is thus right about the object the proof should develop, wrong about the object the solver enumerates. (iii) The flag bound is $12k$, not $\sim 24k$. (iv) The brief's “nondecreasing-vertex-type tie-break” misdescribes the search: the actual invariant is min-type rooting, which is weaker and easier to prove complete (remark after S2). (v) The brief's $k = 1$ statement double-counts the octagon exclusion (Remark 1.5). (vi) 6b should not be proved by importing

Delaney–Dress realization theorems, though the kinship deserves the remark in Section 3; self-contained 2D proofs are shorter than the formal bridge to D-symbols would be.

10 Composition: proof of Theorem 1.6 from the lemmas

Proof of Theorem 1.6, assuming the lemma blocks. (T) is T1.

(S) Let $M \in P_k$. M is a closed configuration built by the DFS, so it is connected (every fresh vertex is glued to the existing component at birth), over $\mathcal{A}$ by construction, and valid because the search vetoed every violation (S4, L1 monotonicity). M passed the rigidity filter, so M is a core (R1). By the C-block (C1, C2, C3a), $\text{dev}(M)$ is an edge-to-edge tiling by unit regular polygons from $\mathcal{P}$, octagon free by Convention 1.3. By R3 its uniformity is exactly the vertex count of M , which is k . Hence $[\text{dev}(M)] \in \mathcal{T}_k$.

(C) Let $[T] \in \mathcal{T}_k$. By B1 and B2a/B2b, $M_0 := T/G(T)$ is a finite connected closed valid core over $\mathcal{A}$ with exactly $k \leq \text{maxnum}$ vertices. By S2 the run roots a search at M_0 's minimal letter; by S1 that search visits a closed configuration $M \cong M_0$; by R1 the filter accepts it; by S3 it is emitted; by P1–P2 the pruner keeps exactly one representative of its isomorphism class, which is in P_k . By B3(a), $\text{dev}(M) \sim \text{dev}(M_0) \sim T$.

(U) Let $M, M' \in P_k$ with $\text{dev}(M) \sim \text{dev}(M')$. Both are cores (R1), so by B3(c) $M \cong T/G(T) \cong M'$ for $T := \text{dev}(M)$. But the pruner never retains two isomorphic solutions: they share a bucket (P2, using A6) and are compared and merged (P1). Hence $M = M'$.

The bijection statement is (S) + (C) + (U). □

11 Lemma ledger

Lemma	Content	Tag	Depends on
D1a	flag calculus, dictionary, face walk	CLOSED	—
D1b	quotients; gluing-type dictionary	CLOSED	D1a, B0
B0	flag rigidity, discreteness	CLOSED	—
O1	octagon species: {4.8.8, 3.8.24}	CLOSED	—
O2	3.8.24 excluded	CLOSED (+cite pin)	O1
O3	octagon forces $t1002$	CLOSED	O1, O2
O4	palette {3, 4, 6, 12}, grid directions	CLOSED (+cite pin)	—
A1	14 configurations, 3 chiral	CLOSED	—
A2	variants = subgroup conjugacy classes	CLOSED	D1a, B0
A3	44 entries, three copies agree	MACHINE (implemented)	A1, A2
A4	structural identities per entry	MACHINE (implemented)	—
A5	ferk transversal / Aut certificate	MACHINE (implemented)	A4
A6	entries pairwise non-isomorphic	MACHINE (added; PASS)	A3
L1	necessity + heredity of conditions	CLOSED	D1a, D1b
L2	mirror parity rule	CLOSED	D1b
T1	termination	CLOSED	—
S1	guided descent (no-drop)	CLOSED	L1, L2, A5, A6
S2	min-letter rooting; sharding partition	CLOSED	S1
S3	emission gates	CLOSED (abstract) + audit	—
S4	picked stub exists; tests = conditions	CLOSED (abstract) + audit	—
R1	refinement = coarsest congruence	CLOSED	—
R2	rejection sound	CLOSED (corollary)	R1, S1–S3, B2, B3
R3	exact uniformity of accepted cores	CLOSED (corollary)	B3
P1	pruner exact on cores	CLOSED	R1
P2	bucket keys are invariants	CLOSED	A6
P3	N cross-check, both directions	MACHINE (PASS, $k \leq 6$)	B3, N-note
C0	quotients preserve valid/ $\angle$	CLOSED	A2(iv)
C1	bundle surface: flat, oriented, closed	CLOSED	C0, A2(iv), L1
C2	closed flat oriented surface = $\mathbb{C}/\Lambda$	CLASSICAL (Killing–Hopf)	—
C3a	developed tiling; grid; rigidity	CLOSED	C1, C2
C3b	affine realization of bundle maps	CLOSED	C3a
C3c	$\mathrm{Fl}(\mathrm{dev}M)/H(M) \cong M$	CLOSED	C3b, B0
C3d	round trip; functoriality	CLOSED	C3b, C3c
C4	<code>eu_develop</code> = marked development	CLOSED + audit	C3a
B1	cocompact wallpaper; periodicity	CLOSED (Bieberbach)	B0
B2a	$T/G(T)$ valid over $\mathcal{A}$, k vtx	CLOSED	D1b, A1, A2, A3, O4, L1, B1
B2b	$T/G(T)$ is a core	CLOSED	C3c, C3d
B3	Galois correspondence; bijection	CLOSED	C3c, C3d, B0

After round 3: every proof row is CLOSED; the MACHINE rows are discharged (A3–A5 audited and re-run, A6 added and passing, P3 passing in both directions for $k \leq 6$); the sole standing CLASSICAL input is C2 (Killing–Hopf). The audit-facing halves of S3, S4, C4 are verified in deliverable B. Deliverable A is complete as written, subject to the referee surface of Remark 8.12.

Round 3 (machine + audit), done: A6 certificate added to the generator; A3/A4/A5 audited against the lemma statements; P3 cross-check wired (`scripts/p3-nkey-crosscheck.ts`); deliverable B written (`audit-deliverable-B.md`); the $k=8$ exhibit root-caused and closed. An adversarial-referee pass (2026-07-12) is folded in: it surfaced no defect in T/S/C/U for the regular palette, and its confirmed minor findings (the O1 two-angle enumeration above, the O3 Step-4 citation above, and the C4 magnitude caveat) are addressed in place.

12 Audit hooks (forward pointers to deliverable B)

The proof is about the abstract algorithm. Its contact points with the C++ are enumerated here so deliverable B can check each; none is discharged yet.

1. **Tables.** A3/A4/A5 certificates: `audit gen_alphabet.py --certify` against the lemma statements; confirm shipped binaries were built from certified tables; add the A6 certificate.
2. **Candidate enumeration** (S1): `extend` tries all free matching-parity stubs including self and mirror, and all letters $g \geq \text{vtype}[0]$ with all `reps`; confirm the starred-stub normalization of `firstfree` loses no candidate.
3. **Rooting and sharding** (S2): `initex` covers every letter; `EU_SHARD_N/W` partitions roots; union-equals-sequential.
4. **Emission** (S3): `seen = 0`; noncounting budget inert on the regular palette; `EU_STREAM` changes the sink only; no other cap in the write path.
5. **Local tests** (S4): `checkpart` against conditions 1/2a/2b; `writecycle/minicycle` free-stub selection.
6. **Pipeline ordering invariant** (R1(iii), P1): `simplify` precedes `comparesolutions` on every path; the two refinement implementations compute the stated fixpoints.
7. **Bucketing** (P2): key is (signature line, fingerprint); buckets never cross k ; `EU_KONLY` semantics.
8. **Developer** (C4): star-walk and BFS guards replaced by (or asserted against) proven bounds; float confined to lattice reduction and coset rounding; the `--sign` convention; area certificate logged as evidence, not proof.
9. **The $k = 8/9$ exhibit:** CLOSED in deliverable B (`audit-deliverable-B.md`, §4, same directory). Root cause: the Python reference solver’s hand-computed attachment bound for the letter $(4, 4, 4, 4)A2$ tried one dart instead of two, missing the starred Aut-orbit; exactly one $k = 8$ tiling (species partition $3+3+1+1$, group $p4m$) is reachable only through a gluing at that dart and was silently dropped. A one-entry fix restores 2850. This is Lemma A5’s failure class realized in the wild, and the A5/A6 certificates are exactly the checks that exclude it from the audited engine.

References

- [1] B. Grünbaum, G. C. Shephard, *Tilings and Patterns*, W. H. Freeman, 1987. Pinned against the copy in `resources/papers`: equation (2.1.2) (the angle equation); Table 2.1.1, pp. 59–61 (17 species, 21 types; the six double-starred species 3.7.42, 3.8.24, 3.9.18, 3.10.15, 4.5.20, 5.5.10 admit no vertex in any edge-to-edge tiling, p. 59); Statement 2.1.3 (the 11 uniform tilings); enantiomorphy of $(3^4.6)$, p. 59.
- [2] L. Bieberbach, Über die Bewegungsgruppen der Euklidischen Räume I, *Math. Ann.* 70 (1911) 297–336. (Cocompact discrete planar groups; any standard crystallographic-groups reference may substitute.)

- [3] J. A. Wolf, *Spaces of Constant Curvature*, 6th ed., AMS Chelsea, 2011. (Space-form classification; used for C2. Any source for Killing–Hopf plus the fixed-point-free classification of planar isometries substitutes.)
- [4] W. P. Thurston, *The Geometry and Topology of Three-Manifolds*, Princeton lecture notes, ch. 13 (orbifolds). (Context only since round 2: the orbifold route was replaced by the direction bundle, Remark in Section 8.)
- [5] A. W. M. Dress, Presentations of discrete groups, acting on simply connected manifolds, *Adv. Math.* 63 (1987) 196–212; A. W. M. Dress, D. Huson, On tilings of the plane, *Geometriae Dedicata* 24 (1987) 295–310. (Context: the D-symbol correspondence; not imported.)
- [6] A. Cardon, M. Crochemore, Partitioning a graph in $O(|A| \log_2 |V|)$, *Theoret. Comput. Sci.* 19 (1982) 85–98; R. Paige, R. E. Tarjan, Three partition refinement algorithms, *SIAM J. Comput.* 16 (1987) 973–989. (Partition refinement computes the coarsest stable partition; used in R1.)
- [7] M. Čtrnáct, tiling solver/pruner algorithm description and implementation, `tools/ctrnact-oracle/reference/` (`algorithm.txt`, `eu_solver.orig.cpp`), 2021–2022.
- [8] TilingAtlas working note, *A canonical form for integer symbols of periodic tilings by regular polygons*, `docs/canonical-form/`, July 2026.
- [9] B. Datta, D. Maity, arXiv:1705.05236; A. K. Kundu, D. Maity, arXiv:2111.15484. (Context only; demoted per Section 9.1.)